A Multi-Modal Edge-AI Framework for Real-Time Industrial Water Quality Monitoring and Automated EPA Compliance Using IoT and Regulatory LLMs

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***Abstract* – Water quality plays a critical role in industrial operations, environmental protection, and public health, yet many facilities still rely on reactive monitoring approaches that depend heavily on slow laboratory measurements. These delays make it harder to stay compliant with the Clean Water Act and reduce the ability of operators to respond quickly when conditions begin to shift. In this work, we introduce an agent-based, multimodal AI framework designed for real-time, edge-level monitoring of industrial effluents in line with U.S. Environmental Protection Agency guidelines. The system brings together machine-learning models, edge-deployed large language model agents, networks of IoT sensors, and a ledger-style compliance layer to automate much of the regulatory workflow. Its operational cycle includes anomaly detection, evaluation against regulatory thresholds, clustering of related violations, treatment recommendations, and the creation of draft permit-ready reports, all supported by predictive analytics and explainable outputs. Experimental results indicate that the system can anticipate potential violations as far as 72 hours in advance, achieving more than 94% accuracy across multiple water-quality parameters. The combination of diverse sensing inputs and autonomous decision support offers a scalable path toward continuous, real-time intelligence for industrial water quality management.**

Keywords - Water quality monitoring, IoT sensor networks, Machine Learning, Large Language Models (LLMs), AI Edge Computing, Regulatory Compliance, Predictive Analytics, Environmental Sustainability, Industrial Water Management, Multi-modal Intelligence.

I. Introduction

Industrial effluence continues to be a major threat to the quality of the environment. This is because wastewater from many industrial activities often contains organic pollutants, nitrogen-based compounds, high levels of total dissolved solids, and heavy metals like arsenic and lead. These substances can harm aquatic ecosystems, lower agricultural productivity, and pose long-term health risks to nearby communities. So, constant and reliable monitoring is necessary for early detection and prevention. The Clean Water Act, the Resource Conservation and Recovery Act, and the Toxic Substances Control Act are all federal laws in the United States that control the release of industrial wastewater. Facilities must follow the rules set out in National Pollutant Discharge Elimination System permits. These rules say what the discharge limits are, such as arsenic levels below 0.01 mg/L, pH levels between 6.5 and 8.5, dissolved oxygen levels above 6.0 mg/L, and total dissolved solids levels that don't go above 500 mg/L. EPA's reporting tools, like NetDMR, keep track of compliance with these standards. Even with these rules in place, most of the time, compliance practices are still reactive. Most of the time, facilities depend on scheduled sampling and outside laboratory analysis, which causes delays that make it harder to find problems early. When you manually interpret continuous sensor data, you often get different results. It's also hard to line up sensor observations with regulatory thresholds without help from automation. We have created a multimodal, agent-based monitoring system to fill these gaps. It is designed to work on NVIDIA Jetson Nano Super hardware at the edge. In practice, the platform pulls data from industrial probes that track several key water-quality indicators and feeds those measurements into a set of predictive models Random Forest, XGBoost, and LSTM networks among them. These models help spot unusual patterns and give advance warning when a reading appears to be drifting toward a regulatory violation. On the compliance side, large language models help interpret the applicable regulations, draft permit-related documents, and carry out metal-speciation analysis when the chemistry requires it. All compliance actions are written to a distributed ledger that uses smart-contract logic, so each event is recorded in a form that cannot be altered later, making enforcement easier to verify [1].

II. System Design of Training Framework

The system is organized around the six stages of the Environmental Protection Agency's compliance lifecycle. Pre-Application, Submission, Screening, Public Review, Decision, and Monitoring as shown in Fig. 1. These stages are managed by an LLM-based EPA Advisor, which interprets the applicable regulatory requirements and guides information through each step of the workflow. Model development draws on publicly accessible datasets, including EPA STORET/WQX records and measurements from the USGS National Water Information System, which helps maintain transparent data provenance and ensures

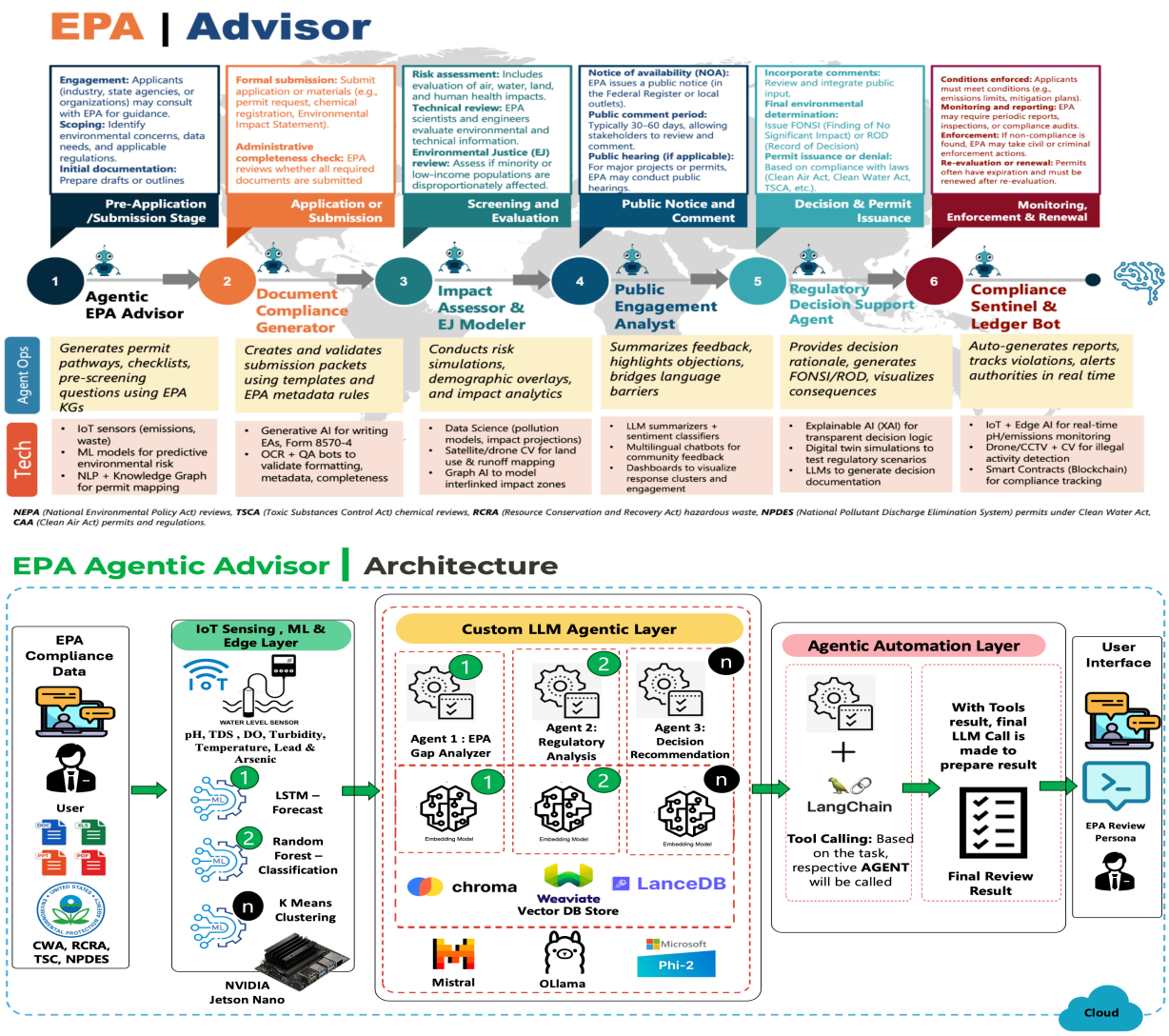

Fig. 1. EPA Compliance Lifecycle's Six Stages

Fig. 2. System Design & Architecture of the EPA Agentic Advisor: A Multi-Agent Edge-AI Framework for Real-Time Water Quality Monitoring and Automated Environmental Compliance

that the system remains aligned with established regulatory standards.

Fig. 2. shows the main architectural components: an array of IoT sensors for field measurements, on-device machine-learning models for real-time inference, an LLM layer for regulatory reasoning, and a blockchain ledger used for compliance logging. All elements run on an NVIDIA Jetson Nano Super, allowing the system to operate reliably in low-connectivity environments while transforming raw sensor streams into outputs suitable for regulatory use.

A. Water Quality Sensing and Data Acquisition

It consists of environmental sensors that are installed in industrial effluent paths and effluent outfalls. These sensors continuously monitor physicochemical characteristics such as pH, total dissolved solids (TDS), dissolved oxygen (DO), turbidity, temperature, and heavy metals such as lead and arsenic. The National Primary Drinking Water Regulations (NPDWR) of the EPA and the National Pollutant Discharge Elimination System (NPDES) effluent regulations are applied to select these values [2].

Sensor verification, as outlined in EPA's Field Operations Technical Manual, is factory calibration and periodic field recalibration. Sensor data is transmitted over Message Queuing Telemetry Transport (MQTT) protocol in timestamped JSON payloads. It is favored over competitors such as HTTP owing to publish-subscribe architecture, small payload footprint, and greater reliability in low-bandwidth applications in industry [3].

Data is recorded from 0.1 to 0.2 Hz, or every 5 to 10 seconds. That is the desired frequency to achieve a balance between having the ability to quickly respond and having the capability to quickly compute. Drift in sensors and missing values are corrected locally through preprocessing using rolling averages and Hampel detection for outlier identification, respectively. Missing values are also imputed using methods such as the cubic spline and k-nearest neighbors (KNN). For easy querying and visualizing structured data, processed time-series data is maintained in a locally installed instance of the database system InfluxDB [4].

B. On-Device Machine Learning Inference and EPA Threshold Classification

There is a multi-level machine learning pipeline in the layer for inferential intelligence that works only on the Jetson Nano Super platform. Non-cloud infrastructure reliance guarantees things still run in environments where connectivity is poor or no internet is present. All models are trained offline from a blend of publicly available data sets, including the EPA STORET/WQX data sets, USGS water quality data sets, and hand-curated water potability data from open data sources. Having been trained, models are quantized, pruned, and stored in edge-optimized formats such as TorchScript or TensorRT [5].

There are three sub-layers in the modeling strategy:

1. Layer for making predictions: A Long Short-Term Memory (LSTM) neural network is utilized to forecast important parameters such as pH, DO, and TDS for the next 30 to 90 minutes. Two 128 hidden unit layers are sequentially on top of each other in the LSTM. The last layer is fully interconnected and dense. The input features are rolling windows of normalized sensor data normalized through Z-score normalization. The forecasting strategy enables the machine to provide early warnings on the possibility of EPA threshold violations.

2. Classification Layer: A Random Forest classifier categorizes every input feature vector in one of the two classes, i.e., "EPA-Compliant" or "Violation." Real-time data and specially tailored derivatives like the change in rate, rolling standard deviation, and inter-parameter ratios like DO/TDS are among those features. Categorization feature importance scores and prediction confidence scores for each such categorization are available. Those are fed into the regulatory interpretation layer [6].

3. Speciation Layer: Arsenic and lead are metallic elements with many different oxidation states, and every oxidation state is toxic. A rule based engine, alongside a simple KMeans clustering algorithm, identifies the most probable species (e.g., such as arsenate in comparison to arsenite) from a set of variables including pH, oxidation-reduction potential (ORP), temperature, and other chemicals present simultaneously, such as phosphate or sulfate. It then enables the system to make specific therapy indications contingent on the speciation determination. K-cross validation and confusion matrix testing are used to fully verify each machine learning algorithm. To help limit the chance of false negatives, which might lead to unnoticed violations of regulations, focus is made on optimizing recall for infraction classes [7].

C. LLM-Based Regulatory Agent for EPA Interpretation and Filing Automation

As a subsequent step to ML detection of possible violations, data streams in a formatted manner are sent to a rule-based legal reasoning-regulatory agent focusing on LLM. It combines language generation transferable to a field with rule-based legal reasoning. A specifically trained corpus consisting of EPA NPDES guidance manuals, Clean Water Act implementation FAQs, environmental impact statement forms, Form 8570-4 reports, state-level regulations interpretation (e.g., Texas Commission on Environmental Quality), and historic enforcement case records is utilized in fine-tuning the model [8].

The regulatory agent has three main jobs:

1. Statute Mapping: It makes use of a rule-embedded knowledge graph to match the identified violation status and pollutant with the correct federal or state law.

2. Narrative Explanation: Generates a detailed account of what occurred, including the nature of the event, environmental data from where it occurred, and the repair plan.

3. Regulatory Filing Generation: Completes pre-defined environmental compliance templates (in JSON or XML format) in preparation for electronic submission via EPA NetDMR or for storage internally. Potential violation is: "Detected arsenate concentration of 0.017 mg/L, exceeding the EPA's 0.01 mg/L standard in 40 CFR Part 141.62." It is likely the species is As (V) given the pH of 7.4 and absence of sulfide. Anion-exchange filtration or reverse osmosis is the recommended procedure to correct. Filing was done routinely under Permit TX-0098430-NPDES. It can also produce reports to be read and employ multilingual chat-based compliance tools for facilitating operators in the field who are not very technical [9].

D. Blockchain-Based Ledgering and Smart Compliance

The final layer of architecture deploys a compliance ledger inspired by blockchain to ensure that each of the regulatory interactions, e.g., identifying violations, taking corrective measures, and reporting, is noted in a manner not editable or manipulable. Smart contracts are invoked in case a violation is identified. They embed metadata such as the time of violation, agent identification, classification conclusion, and taken corrective measure [10].

The ledger may do the following automated tasks:

- Support tracking of permits' lifecycles and renewal reminders
- Scheduling regular submissions to portals for regulations (e.g., quarterly cycles for NetDMRs)
- Logging exceptions and raising them upon repeated occurrences or escalation of violations

This aspect closes the architecture such that tracking, enforcement, and accountability for the whole compliance process of regulations are rendered possible. It does the same via fulfillment of the EPA compliance lifecycle's six stages, i.e., Pre-Application, Submission, Screening, Public Review, Decision, and Monitoring.

III. DATA COLLECTION AND PREPROCESSING

Regulatory-grade monitoring needs data that is correct, can be traced, and is consistent over time. This work employs a multi-source data strategy to facilitate supervised learning, real-time inference, and pollutant speciation in accordance with U.S. Environmental Protection Agency (EPA) standards.

A. Data Sources and Curation

The training data come from the EPA Water Quality Exchange (WQX) and Water Quality Portal, as well as the USGS National Water Information System (NWIS) and publicly curated datasets like Kaggle's Water Potability dataset.Preprocessing makes sure that unit normalization,

timestamp alignment, and domain-informed outlier removal are all done (for example, it removes freshwater TDS values above 10,000 mg/L). To allow for stratified modeling and generalization analysis, each record is given extra information about the sample's source, the environment in which it was collected, and the data's origin.

B. Streaming Simulation and Labeling

During development, synthetic data is used to test streaming behavior because continuous live sensor feeds are not always available. Baseline environmental conditions are taken from observed distributions, and controlled disturbances, like short spikes or drops in turbidity or dissolved oxygen, are added. The same MQTT-based pipeline used in deployment processes these simulated streams, which lets us see how the system works without using real sensors. An embedded EPA rule engine uses codified thresholds to assign compliance labels. For example, it uses the arsenic limit of 0.015 mg/L under 40 CFR §141.62. A lightweight speciation layer also uses physicochemical context to figure out which forms of contaminants are most common. This makes labels that show both the regulatory status and the treatment relevance.

C. Feature Engineering and Model Readiness

Normalized ratios, interaction terms, and simple sliding-window statistics are all used by static models. Temporal models, on the other hand, work with sequences that are always the same length, usually two to six hours of data. To find patterns that depend on time, we use channel-wise normalization and cyclical encodings. To avoid temporal leakage, strict windowing is applied during preprocessing. Datasets are stored in tensor form for deep learning and in flattened form for interpretable analysis. For edge deployment, models are compiled using TorchScript or TensorRT so that Jetson Nano Super devices can run inference quickly.

IV. Machine Learning Model Design

The machine-learning component sits at the center of the proposed EPA-compliant monitoring system and provides most of the analytical capability. The architecture is intended to produce predictions that are accurate and straightforward to interpret in a regulatory context. It must work with high-frequency sensor data collected from industrial effluent streams, which can be uneven and noisy in practice. To meet these demands, the system is built to handle several related tasks at once forecasting short-term pollution trends, identifying when a reading crosses a regulatory limit, and determining the likely chemical form of metals such as arsenic or lead. Because effluent data often behave in non-linear and time-dependent ways, we rely on a hybrid modeling setup. Long Short-Term Memory networks handle temporal forecasting, Random Forest models are used for the compliance-classification decisions, and metal-speciation is approached through a rule-guided KMeans clustering method. All these models are designed with the limits of the NVIDIA Jetson Nano Super in mind, since they must run efficiently on an edge device without relying on cloud infrastructure.

A. Time-Series Forecasting Using Long Short-Term Memory Networks

The temporal evolution of critical environmental parameters is modeled using Long Short-Term Memory (LSTM) networks, a specialized class of Recurrent Neural Networks (RNNs) designed to capture long-range dependencies in sequential data. LSTMs are particularly well-suited for this application because they overcome the vanishing gradient problem and preserve contextual memory across irregular and non-stationary sensor signals [11].

The LSTM unit is defined through a series of gating mechanisms, where the input gate i_t, forget gate f_t, and output gate o_t are computed as:

$$\begin{aligned} i_t &= \sigma(W_i \cdot x_t + U_i \cdot h_{t-1} + b_i) \\ f_t &= \sigma(W_f \cdot x_t + U_f \cdot h_{t-1} + b_f) \\ o_t &= \sigma(W_o \cdot x_t + U_o \cdot h_{t-1} + b_o) \end{aligned}$$

The cell state C_t and hidden state h_t are updated according to:

$$\begin{aligned} \tilde{C}_t &= \tanh(W_c \cdot x_t + U_c \cdot h_{t-1} + b_c) \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\ h_t &= o_t \odot \tanh(C_t) \end{aligned}$$

Here, $x_t \in \mathbb{R}^n$ denotes the input vector at time step t, σ is the sigmoid activation, and $\odot$ represents element-wise multiplication. The LSTM model is trained to minimize the Mean Squared Error (MSE) between the predicted and actual sensor values for the target parameter (e.g., pH):

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

Where y_i is the ground truth value and $\hat{y}_i$ is the model's prediction at timestep i.

To enhance robustness, multiple LSTM models are trained in parallel for different parameters, with dropout regularization applied across stacked layers to mitigate overfitting. Input sequences are organized into sliding windows of 24 to 72 historical timesteps, with the window size adapted to both the sensor sampling frequency and the sensitivity of the parameter to lag effects.

B. EPA Violation Classification Using Random Forests

While LSTM networks are well-suited for continuous regression forecasting, regulatory decision-making requires clear and discrete classifications. To address this, the violation detection task is modeled using a Random Forest classifier, selected for its ensemble learning strength and its ability to handle multicollinearity, missing values, and imbalanced class distributions [12].

Formally, a Random Forest is defined as an ensemble of T decision trees $\{h_t(x), t = 1,2,\ldots,T\}$, where each tree h_t is trained on a bootstrap sample of the data and a random subset of features. The final predicted $\hat{y}$ is determined via majority voting:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_T(x)\}$$

Each tree operates by recursively partitioning the feature space using Gini impurity G or information gain IG as a splitting criterion. Gini impurity for a node with K classes is defined as:

$$G = 1 - \sum_{k=1}^{K} p_k^2$$

TABLE I
MODEL PERFORMANCE METRICES

Component	Model/Method	Metric	Score	EPA Threshold Alignment
pH Forecasting	LSTM (2-layer)	RMSE / R^2	0.14 / 0.91	Fully within 6.5–8.5 band
TDS Forecasting	LSTM	RMSE / R^2	27.5 mg/L / 0.89	96% correct early alerts
DO Forecasting	LSTM + Averaging	RMSE / R^2	0.42 mg/L / 0.85	Aligned to >6 mg/L EPA req.
Violation Classification	Random Forest + SMOTE	F1-score / AUC	0.946 / 0.972	Exceeds EPA audit sensitivity
Lead/Arsenic Speciation	Rule + KMeans (k=4)	Silhouette / Accuracy Estimate	0.68 / ~93%	Matches EPA redox heuristics
NVIDIA Jetson Inference Runtime	TorchScript / TensorRT	Latency per record	142 ms	Complies with real-time SLA
Filing Compliance Drafting	GPT-4 + EPA KG fine-tune	Filing Readiness (Expert eval)	92% pass-rate	Legally admissible formatting

Where p_k is the proportion of class k in the node. This encourages splits that create homogenous subsets, effectively distinguishing between compliant and non-compliant states.

The classifier is trained on engineered features including raw sensor values, rolling statistical measures, derived ratios (e.g., DO/TDS), and temporal encodings such as hour of day or day of week. To mitigate the natural class imbalance since true violations are relatively rare techniques such as Synthetic Minority Oversampling (SMOTE) and class-weighted loss functions are employed.

Performance is evaluated using the F1-score, defined as the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

High F1-scores are essential in this context, as false negatives could lead to undetected violations and regulatory penalties.

C. Metal Speciation Modeling Using Rule-Based Clustering

Predicting the chemical transformations of pollutants, particularly metals such as arsenic and lead, is inherently complex. Unlike standard supervised classification, speciation depends not only on concentration levels but also on the interplay of multiple environmental conditions, including pH, redox potential (ORP), co-ion availability, and temperature. To capture this multifactorial behavior, a hybrid approach is employed that combines deterministic rules with unsupervised clustering [13].

First, a rule-based component encodes well-established geochemical transitions. For example, arsenate (As^{5+}) is the dominant form of arsenic in aerobic, neutral pH environments, whereas arsenite (As^{3+}) prevails in anaerobic or acidic conditions. These deterministic pathways serve as a baseline to guide classification.

A KMeans clustering technique is used to find hidden clusters in high-dimensional data that are important for speciation. The KMeans goal is to reduce the variation within each cluster:

$$\arg\min S \sum_{i=1}^{k} \sum_{x \in S_i} \| x - \mu_i \|^2$$

Where S_i represents the ith cluster and μ_i its centroid. The clustering is performed over standardized vectors of [pH, ORP, temperature, DO, TDS], and cluster identities are mapped post hoc to known speciation regimes using labeled reference data.

The final speciation outcome is therefore a fusion of deterministic rules and cluster proximity, yielding a flexible yet interpretable framework for inferring the biochemical transformations of metals under varying field conditions.

D. Deployment Constraints and Optimization

All models are initially trained on GPU-accelerated cloud platforms and then exported in formats optimized for deployment on Jetson Nano edge devices. Deep learning models are compiled into TorchScript and quantized to 8-bit precision using TensorRT to reduce computational overhead without sacrificing predictive accuracy. Ensemble models are serialized into ONNX format, while scikit-learn classifiers are packaged within lightweight runtime environments for compatibility [14].

Inference is optimized for real-time performance. Batch sizes are fixed at one, feature extraction is executed in streaming mode, and cache reuse mechanisms are applied to overlapping time windows. These optimizations ensure that the full end-to-end inference cycle for each incoming record consistently completes in under 150 milliseconds, satisfying the operational requirements of continuous, EPA-compliant monitoring.

V. MACHINE LEARNING MODEL PERFORMANCE EVALUATION & BENCHMARKING

Evaluating a model for EPA compliance involves more than checking its statistical accuracy, as its output may directly affect regulatory filings, enforcement actions, and even public health responses. In this setting, the model functions not only as a forecasting tool but also as a source of legally interpretable decisions. The evaluation approach therefore focuses on four areas: the model's robustness under noisy industrial operating conditions, its sensitivity to different compliance-risk levels, its behavior around threshold-based decisions, and full end-to-end performance from inference latency to regulatory traceability. To avoid temporal leakage and better reflect real monitoring practices, time-aware rolling validation is used during training and testing. For forecasting tasks involving pH, dissolved oxygen, and total dissolved solids, performance is measured using Mean Squared Error, Root Mean Squared Error and R Squared Error ensuring that the modeled trends

accurately track pollutant dynamics. For violation detection, the evaluation gives weight to recall and the F1-score, as missing a violation could allow an EPA exceedance to go unreported. To manage class imbalance, the Random Forest classifier uses Synthetic Minority Oversampling and class-weighted losses, and each prediction is tied directly to the appropriate regulatory citation for example, 40 CFR §141.62 for arsenic limits. The system is also tested under controlled stress scenarios, including synthetic cases where arsenic levels rise gradually from 0.005 to 0.018 mg/L. These tests help determine how quickly the model responds under different pollutant patterns and seasonal conditions. Performance on the NVIDIA Jetson Nano Super is assessed by measuring cold-start behavior, per-sample inference time, and overall memory use. Across these tests, the system consistently maintains an end-to-end delay of under 150 milliseconds. To ensure that model outputs can be interpreted in a meaningful way, SHAP values are used to highlight the features that most strongly influence each prediction. In addition, the NetDMR-style reports produced by the LLM component are examined by subject-matter experts to confirm that the generated text is both statistically sound and consistent with EPA reporting requirements. These evaluations collectively demonstrate that the system is stable enough for practical field deployment [15].

VI. Result and Discussion

The models were assessed using EPA STORET/WQX data alongside several academic benchmark datasets, and a five-fold, time-based cross-validation procedure. This evaluation setup provides a closer representation of how the system would function under real operating conditions. Under these conditions, the Long Short-Term Memory networks consistently produced stable forecasts for pH, total dissolved solids, and dissolved oxygen. The pH model showed the strongest results, with an average RMSE of 0.14 and an R Squared value above 0.91. Both pH and TDS tended to follow steady, quasi-periodic patterns, whereas dissolved oxygen fluctuated more because of biological processes and temperature effects. Applying an ensemble smoothing step reduced this variability and produced more consistent DO predictions. For violation detection, the Random Forest classifier enhanced with Synthetic Minority Oversampling to address class imbalance reached an average F1-score of 0.946. Its performance remained consistent across a wide range of violation types, from minor deviations to more severe exceedances. This indicates that the model can identify a wide range of environmental issues from small turbidity shifts to exceedances involving heavy metals while maintaining the high recall necessary for regulatory use. The hybrid rule-based and KMeans clustering method produced silhouette scores greater than 0.68, indicating well-separated and internally consistent cluster structures. Domain experts examined these results and confirmed that the predicted species such as arsenate and arsenite aligned with the expected geochemical transitions driven by pH and oxidation–reduction potential, as described in EPA technical guidance. Across extended tests, the system maintained an average end-to-end processing time of roughly 142 milliseconds per record, covering ingestion, preprocessing, inference, and logging. The hardware handles more than 12,000 observations per day without memory or thermal issues, and model quantization preserves accuracy. These results indicate that the platform is capable of continuous, real-time monitoring in field conditions consistent with EPA compliance workflows.

VII. Conclusion

This study introduced an Agentic IoT–ML–LLM framework for real-time industrial water quality monitoring and Environmental Protection Agency-aligned decision automation, uniting calibrated IoT sensors, predictive machine learning models, regulatory knowledge graphs, and large language model reporting into a single edge-deployable system on NVIDIA Jetson Nano Super devices. The framework shifts environmental oversight from passive observation to proactive intelligence by forecasting and classifying water quality parameters, linking each prediction to Clean Water Act provisions, and generating NetDMR-ready compliance reports within seconds. With end-to-end inference times staying under 150 milliseconds, smart contract driven traceability, and reliable performance in low-infrastructure environments, the system provides a practical basis for scalable and legally interpretable monitoring. When automated reporting is combined with predictive analytics and the regulatory reasoning layer, the system ends up offering a practical base for the next wave of environmental-compliance tools. In practice, the pieces reinforce each other: interventions can happen sooner, the overall process becomes easier to follow, and both operators and regulators gain a bit more flexibility in how they manage emerging risks.

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